FAISHAL YUSUF BAQIR

Software Developer Project Officer · Flutter Mobile Developer

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SUMMARY

Software Developer Project Officer with 2 years of experience building offline-first Flutter apps for palm oil plantations: harvest, inspection and attendance. Leading the mobile side of DO! IAM, a new plantation information management system for 3 palm oil groups, built from scratch. Hands-on with on-device ML (TensorFlow Lite face recognition, liveness, YOLO object detection), NFC, GPS geofencing, Bluetooth thermal printing and SQLite. Telecommunication Engineering graduate from Telkom University with an IEEE publication and Bangkit Academy 2023 background.

SKILLS

- **Mobile:** Flutter, Dart, Java (Android), GetX, Clean Architecture
- **Data & offline:** SQLite / Drift, offline-first sync, REST API (Dio)
- **On-device ML:** TensorFlow Lite, Google ML Kit, YOLO, face recognition and liveness detection
- **Hardware & geo:** NFC, GPS / geofence, Bluetooth thermal printer, offline maps (PMTiles), GIS routing
- **Cloud & tools:** Firebase, Google Cloud Platform, Git (GitHub, AWS CodeCommit), Postman, Play Store release

EXPERIENCE

PT Lentera Kreasi Teknologi (part of First Resources Ltd) · Jakarta

Software Developer Project Officer — Nov 2025 – Present

- Lead mobile development of DO! IAM, a new plantation information management system built from scratch for 3 palm oil groups (FR, FAP, KAS), delivered in 2 phases: NFC, then Bluetooth thermal printer.
- Plan and deliver offline-first features used daily by harvesters, supervisors and field assistants: harvest tickets, SPB, fruit quality, seedling receipt and 5 types of field inspection with a verification flow.
- Replaced FaceNet-512 (~47 MB) with an ArcFace-based MobileFaceNet (6.9 MB) and added face alignment for attendance; verification TAR at FAR 1e-4 rose from 84.6% to 99.1% on the LFW benchmark.
- Added on-device passive liveness (anti-spoofing) and an enrollment focus check to stop photo spoofing and blurry face registrations.
- Built on-device PPE detection (YOLO, 6 classes, mAP50 0.83), from model training to Flutter integration in a background isolate.
- Built an offline aerial map (PMTiles) with route recording, field findings, a ticket workflow (Open → Progress → Completed → Closed) and offline road routing (Dijkstra on a GIS shapefile) with an online fallback.
- Fixed memory issues on low-end field devices: a TFLite native leak (~60 MB per scan), a 170 MB map asset reload, and an inspection loader that now loads 10,000 records in ~0.5 s.
- Integrated Hikvision face data sync, GPS geofencing, NFC read/write and Bluetooth thermal printing.

Programmer Enterprise System 1 — Sep 2025 – Nov 2025

- Joined after the ANJ–First Resources acquisition as IT consultant for the group companies.
- Built face recognition attendance with camera and gallery enrollment, and QR scanning of TPH (harvest collection points) within a 100 m geofence.
- Maintained ePCS, a native Java plantation control app that reads TPH data over NFC.

PT Austindo Nusantara Jaya Tbk (ANJ) · South Jakarta

Mobile Apps Developer Staff — Dec 2024 – Aug 2025

- Developed and maintained ePMS (plantation, harvest and inspection) and eTis (external harvest orders), both in Flutter with NFC and Firebase notifications.
- Refactored data sync and local database fetching to cut sync time.
- Maintained and released 2 apps on the Play Store, including eTrace.

Adaptive Network Laboratory, Telkom University · Bandung

Research Assistant Jun 2022 – Sep 2024

- Selected as 1 of about 20 research assistants from 100 candidates.
- Researched next-generation software-defined networking with P4 (Programmable Protocol-Independent Packet Processors).
- Managed NGSDN instances on the lab's OpenStack server. Tools: Linux, Ryu, ONOS, P4, OpenStack.

Bangkit Academy 2023 (led by Google, GoTo, Traveloka) · Bandung

Cloud Computing Cohort Feb 2023 – Jul 2023

- Completed 7 cloud computing courses across Dicoding, Coursera and Google Cloud Skills Boost. Certificate
- Built the backend API (Node.js, Express) for the capstone app Cycle-Me and deployed it on GCP (App Engine, Cloud Run, Cloud SQL, Compute Engine).

PROJECTS

- **WasteApp — Final Capstone, Telkom University (2023–2024):** Flutter app for a Bandung waste bank that detects waste types with a YOLOv8 model through a REST API; Firebase Auth and Firestore backend. Released on the Play Store, registered as intellectual property (HKI) and published in IEEE. GitHub · Team repos
- **FlixID (2024):** Movie ticket app built in a Udemy Flutter course, using the TMDB API, Firebase, Riverpod and clean architecture. GitHub
- **My-WB (2023):** Flutter app for the Dicoding final submission that detects waste and recommends reuse, reduce and recycle videos. GitHub
- **Cycle-Me (2023):** Bangkit capstone backend with Node.js, Express and a Python Flask ML service, deployed on GCP by a team of 6. GitHub
- **Posyandu Alamanda website (2022):** Community-service website for a health post in Gedebage, Bandung, with registration for mother and child services. Site

EDUCATION

Telkom University · Bandung Sep 2020 – Aug 2024

Bachelor of Engineering, Telecommunication Engineering · GPA 3.54 / 4.00 · MSIB graduate (Bangkit Academy 2023)

PUBLICATIONS & CERTIFICATIONS

- *Advanced Waste Detection: MobileNetV2 SSD and YOLOv8 for Waste Bank Mobile Application* — IEEE, 2024. IEEE Xplore
- *WasteApp: Mobile-Based Waste Bank Application with Object Detection Method on YOLOv8* — HKI (intellectual property registration), 2024. Certificate
- *Digitalisasi Pusat Informasi dan Administrasi Data pada Posyandu Alamanda Berbasis Web* — Open Library Telkom University, 2022. Article
- **Certifications** (Google Cloud Skills Boost profile): Google Cloud Computing Foundations: Data, ML, and AI in Google Cloud · Preparing for Your Associate Cloud Engineer Journey · Perform Foundational Infrastructure Tasks in Google Cloud · Flutter Basic (Dicoding) · Belajar Dasar Pemrograman JavaScript (Dicoding)

ORGANIZATION

- **People Development, Adaptive Network Laboratory** (Jul 2022 – May 2024): ran monthly NetClass tech talks and NetNongkrong team events for lab assistants; supported the lab's profile video (2022).
- **Community service, Posyandu Alamanda, Gedebage** (2022): built and handed over the health post's website.